

Bitcoin Market Sentiment vs Trader Performance Analysis

Submitted By

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Technologies Used

Python, Pandas, NumPy, Matplotlib, Seaborn, Google Colab

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1. Introduction

Cryptocurrency markets are highly volatile and heavily influenced by trader emotions such as fear and greed. Market sentiment plays a major role in determining trading activity, investment decisions, and overall market trends.

This project analyzes the relationship between Bitcoin market sentiment and trader performance using historical trading data and the Bitcoin Fear and Greed Index dataset.

The goal of this analysis is to understand how trader profitability, trade size, and trading behavior change under different emotional market conditions.

2. Objective

The objectives of this project are:

- To analyze trader behavior during Fear and Greed market conditions
- To compare trader profitability across different sentiments
- To study trading activity and trade size patterns
- To identify high-performing traders
- To derive insights for improving trading strategies

3. Datasets Used

3.1. Bitcoin Fear and Greed Dataset

This dataset contains daily Bitcoin market sentiment classifications.

Important Columns

- Date
- Classification

3.2. Historical Trader Dataset

This dataset contains historical trade records from Hyperliquid.

Important Columns

- Account
- Coin
- Execution Price
- Size USD
- Side
- Timestamp IST
- Closed PnL
- Fee

4. Tools and Technologies

Tool / Library	Purpose
Python	Data analysis and preprocessing
Pandas	Data manipulation
NumPy	Numerical operations
Matplotlib	Visualization
Seaborn	Statistical charts
Google Colab	Notebook environment
GitHub	Project hosting

Table 1: Tools and Technologies

5. Data Preprocessing

Before performing analysis, both datasets were cleaned and processed.

The historical trading dataset contains thousands of trading records collected from Hyperliquid, while the Fear and Greed dataset contains daily Bitcoin market sentiment classifications.

5.1. Preprocessing Steps

- Checked missing values
- Removed duplicate records
- Converted date columns into datetime format
- Created a common date column
- Merged datasets using the date field
- Created additional calculated columns

6. Methodology

The project followed a structured data analysis workflow involving data collection, preprocessing, merging, exploratory data analysis, visualization, and insight generation.

Both datasets were merged using a common date field to analyze trader behavior under different market sentiment conditions.

7. Exploratory Data Analysis

Exploratory Data Analysis was performed to identify important patterns and trends.

7.1. Market Sentiment Distribution

This chart shows the distribution of Fear and Greed market conditions.

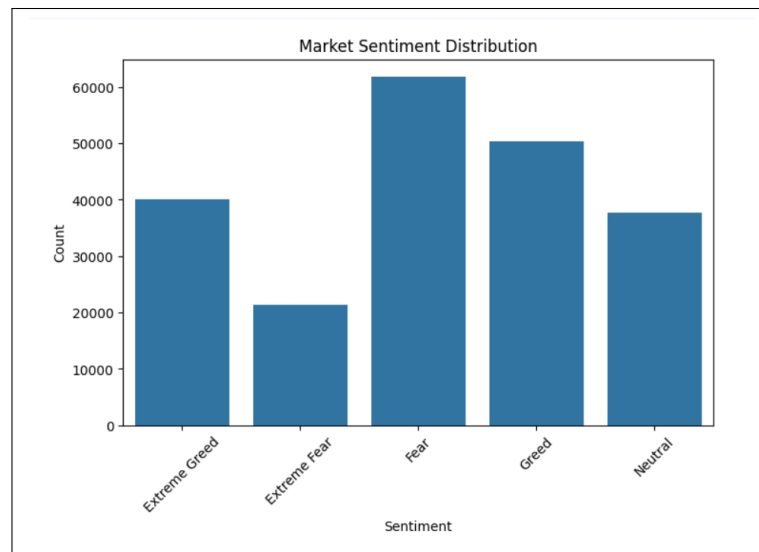


Figure 1: Market Sentiment Distribution

7.2. Average Profit Analysis

This analysis compares average trader profit during different market sentiments.

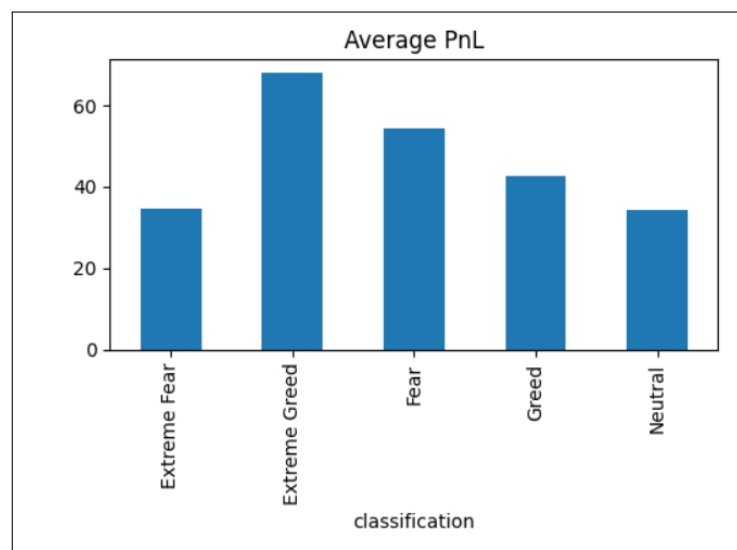


Figure 2: Average Profit by Sentiment

7.3. Total Profit Analysis

This chart represents total trader profit across different sentiment categories.

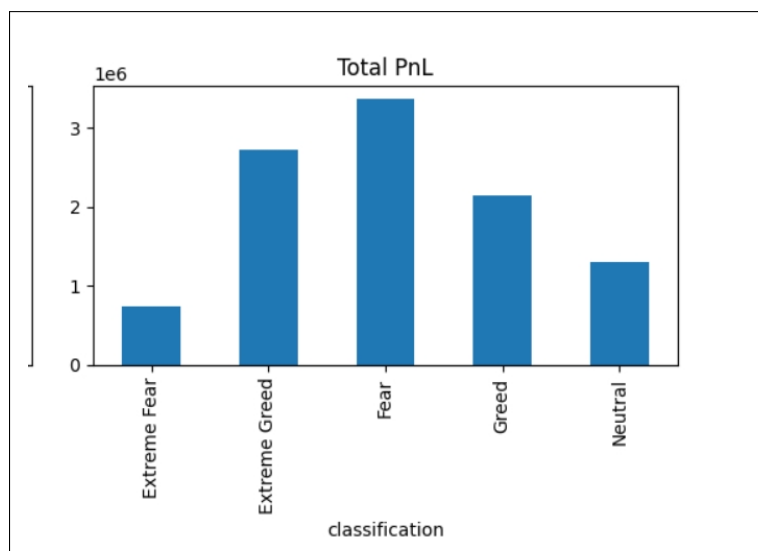


Figure 3: Total Profit by Sentiment

7.4. Trading Activity Analysis

This graph represents daily trading activity across the dataset.

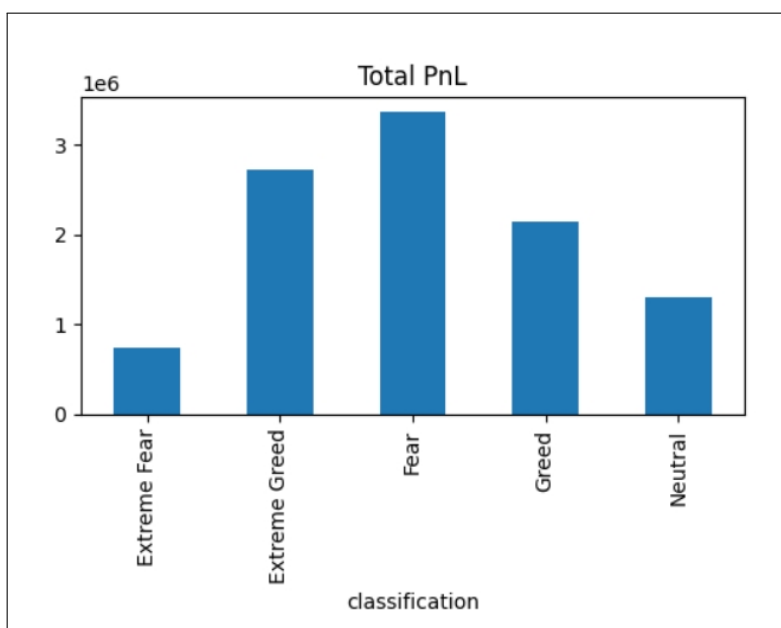


Figure 4: Daily Trading Activity

7.5. Trade Size Analysis

Trade size analysis helps understand trader risk exposure.

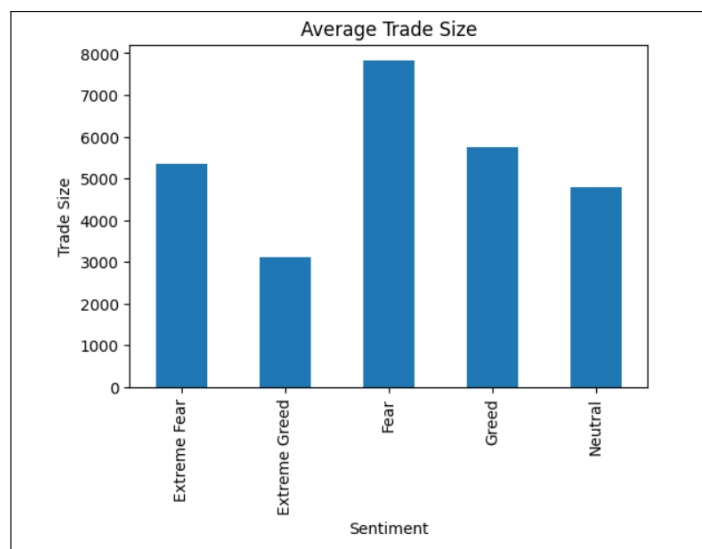


Figure 5: Average Trade Size by Sentiment

7.6. Correlation Analysis

Correlation analysis identifies relationships between important trading variables.

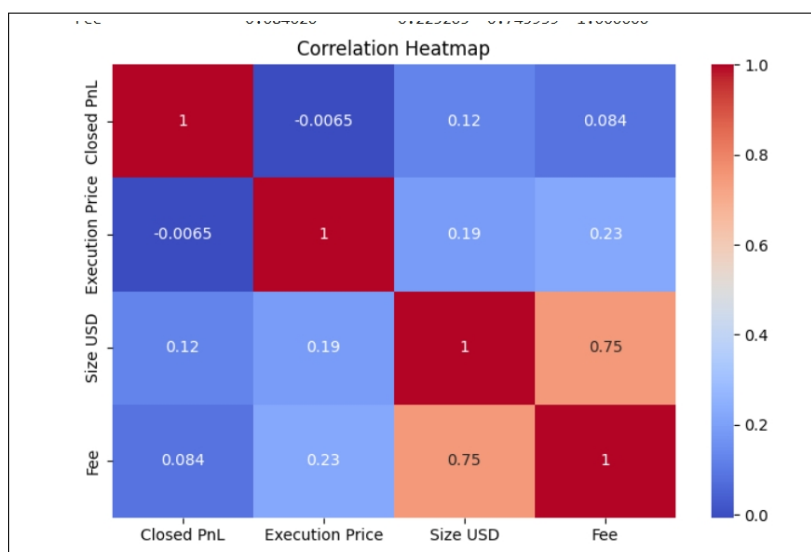


Figure 6: Correlation Heatmap

8. Key Insights

The following insights were identified during analysis:

- Market sentiment strongly influences trader behavior.
- Greed periods show higher trading activity and increased trade size.
- Fear periods reflect cautious market participation.
- Certain traders consistently outperform others.

- Larger trade sizes increase both profit potential and trading risk.
- Emotional market conditions impact trader decision making.

9. Business Impact

The findings from this project can help improve:

- Trading strategy development
- Sentiment-based market prediction
- Risk management systems
- Trader behavior analysis
- Financial decision support systems

10. Challenges Faced

Some challenges encountered during the project include:

- Cleaning large datasets
- Handling date conversion issues
- Merging datasets with different structures
- Interpreting emotional market behavior

11. Future Improvements

This project can be extended further by:

- Applying Machine Learning models
- Building real-time dashboards
- Performing live sentiment tracking
- Adding more cryptocurrency datasets
- Using advanced predictive analytics

12. Conclusion

This project successfully analyzed the relationship between Bitcoin market sentiment and trader performance.

The analysis showed that Fear and Greed significantly influence trading behavior, profitability, trade size, and market participation.

Greed periods generally demonstrated higher trading activity and larger trades, while Fear periods showed cautious trading behavior.

The generated insights can help improve trading strategies, market analysis systems, and risk management approaches in cryptocurrency markets.

The project demonstrates how sentiment-driven analytics can support smarter trading decisions and improve market understanding in cryptocurrency trading environments.

13. References

- [Bitcoin Fear and Greed Dataset](#)
- [Hyperliquid Historical Trader Dataset](#)
- [Python Documentation](#)
- [Pandas Documentation](#)
- [Matplotlib Documentation](#)
- [Seaborn Documentation](#)